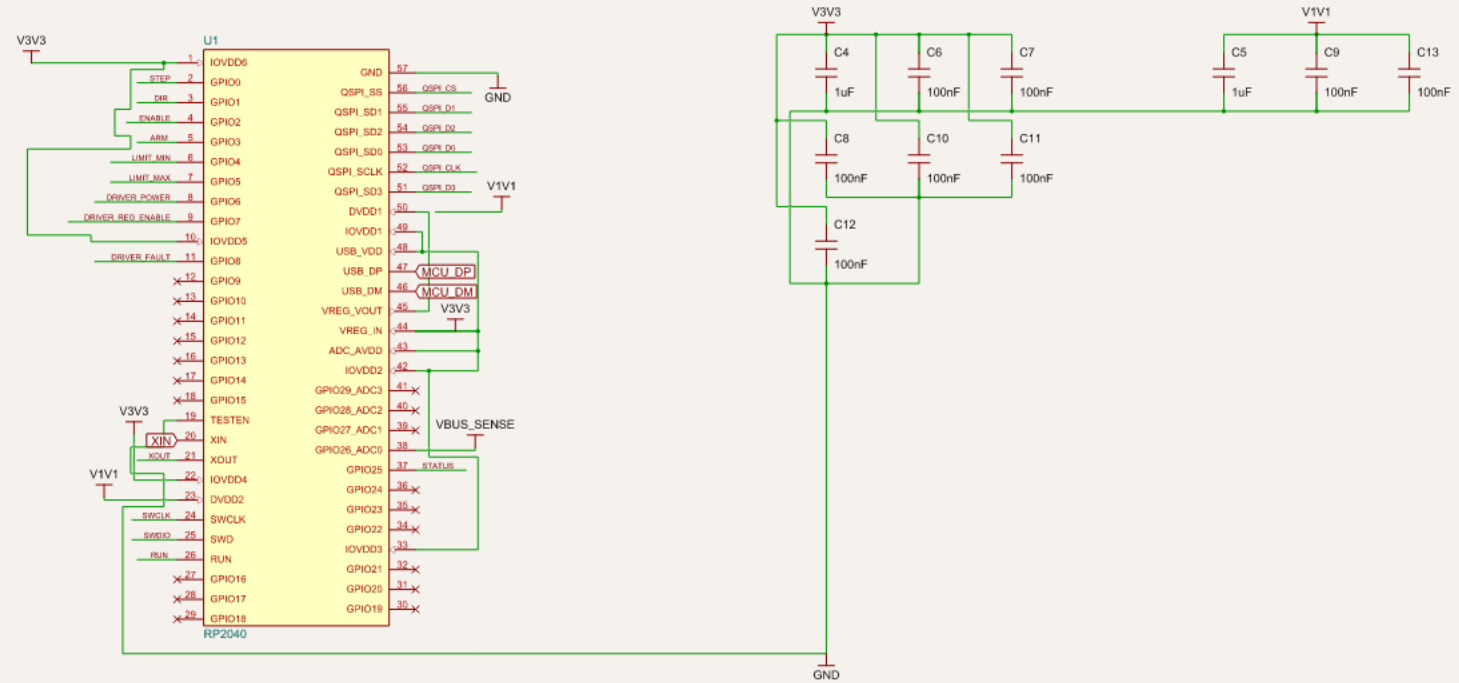


U1 / RP2040 motion + USB controller
 3.3 V I/O; internal 1.1 V core rail
 12 MHz crystal; firmware limit 2000 steps/s



1

2

3

4

5

6

A

A

B

B

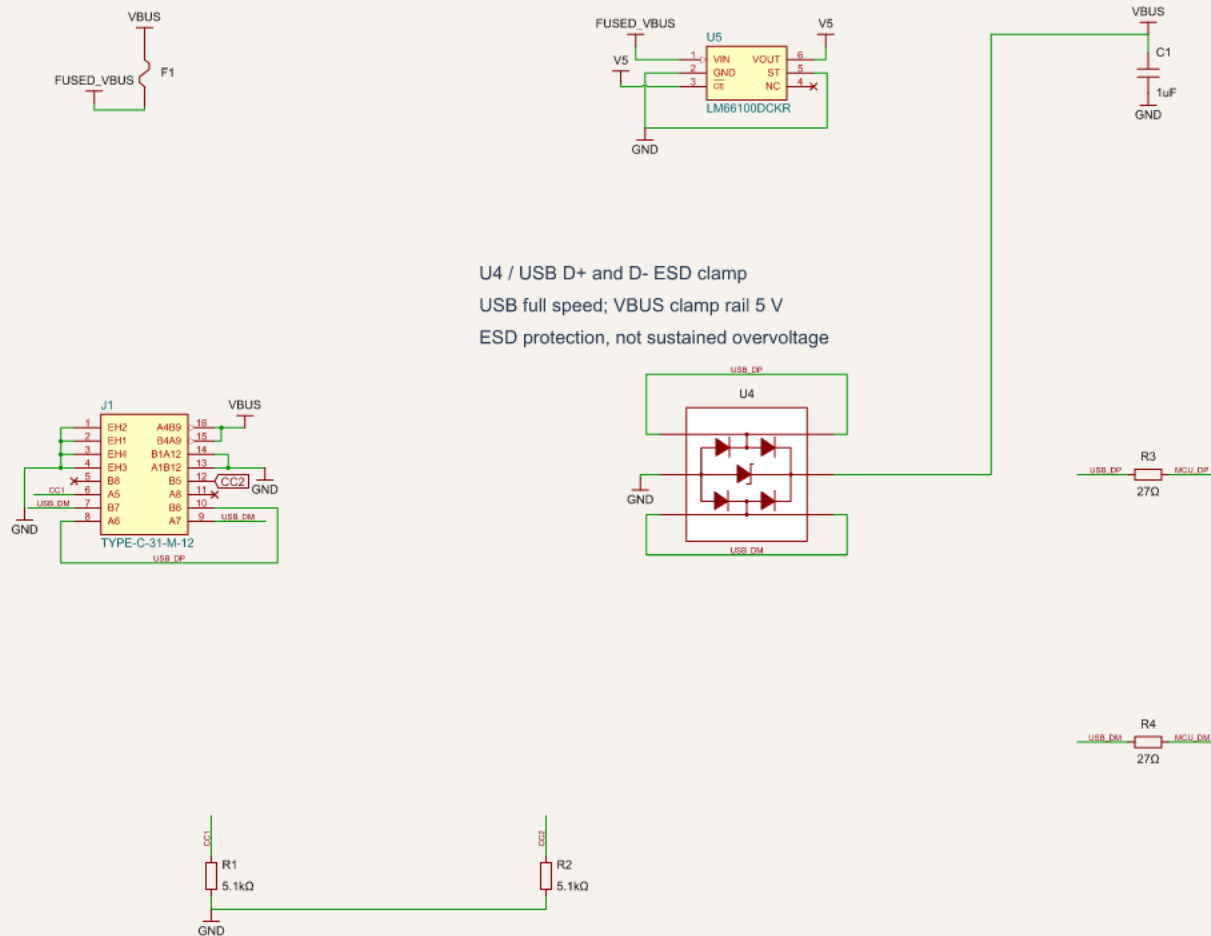
C

C

D

D

U5 / reverse-current blocking diode
5 V USB path; IC rated 1.5 A
Board USB request: 500 mA after config



1

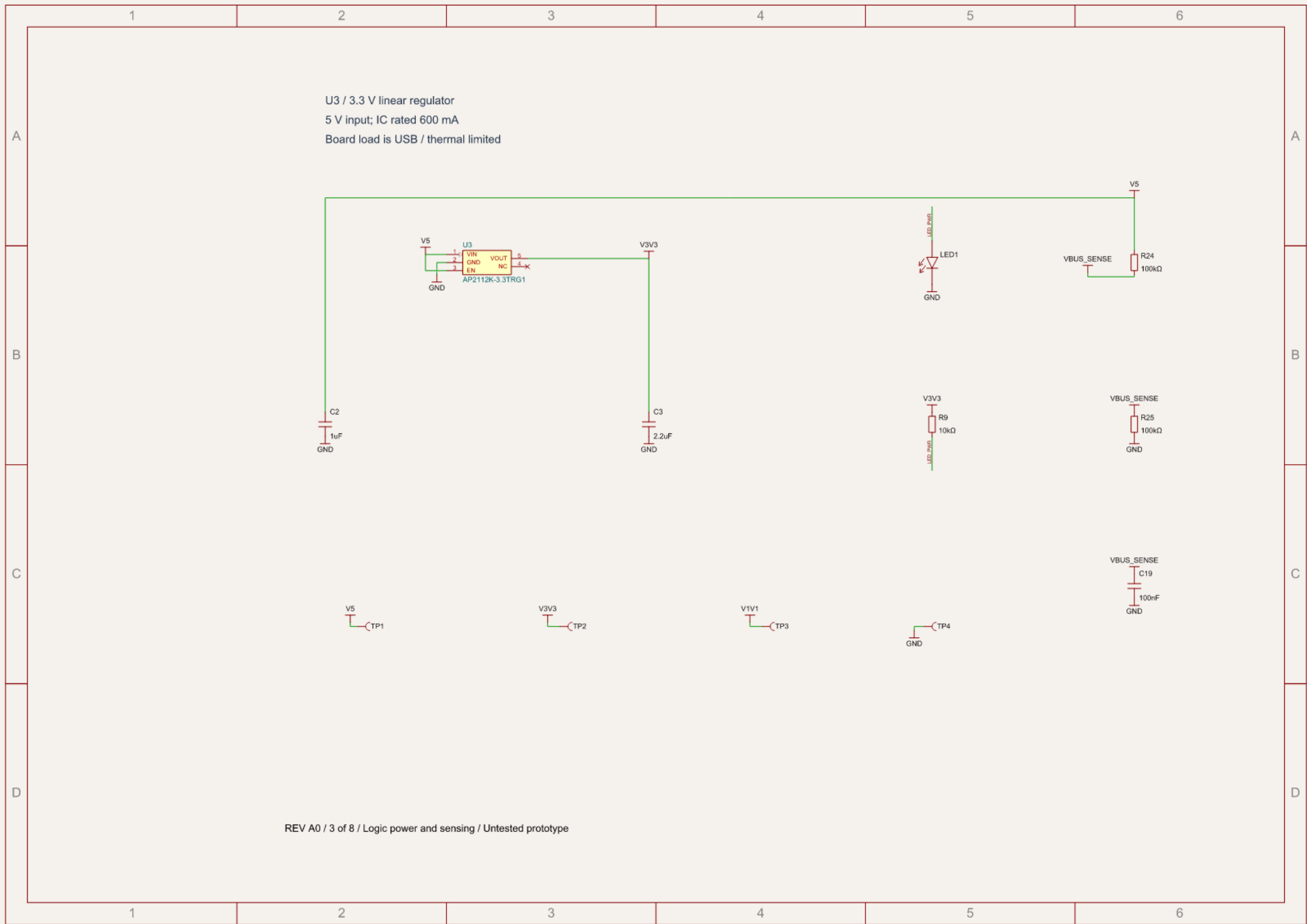
2

3

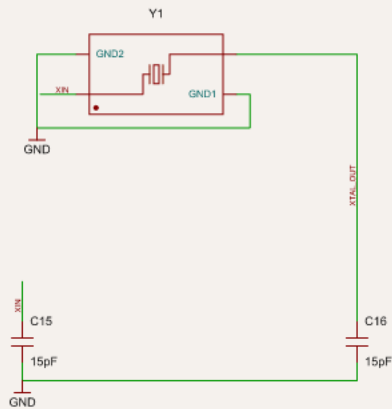
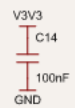
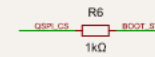
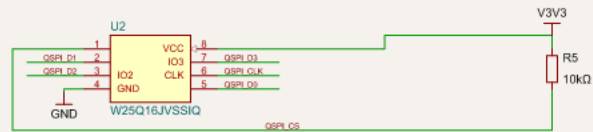
4

5

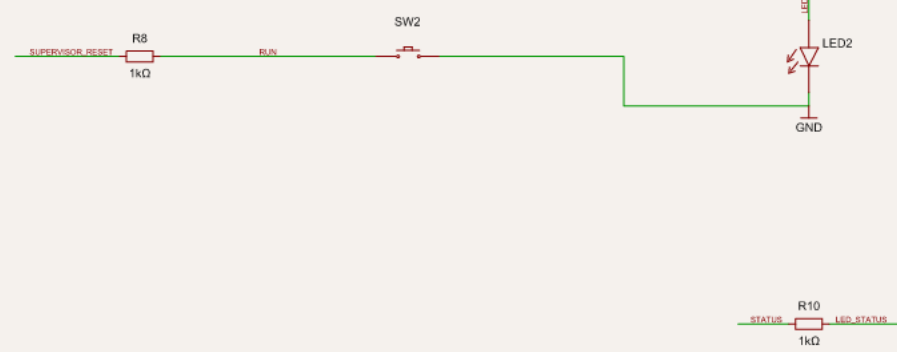
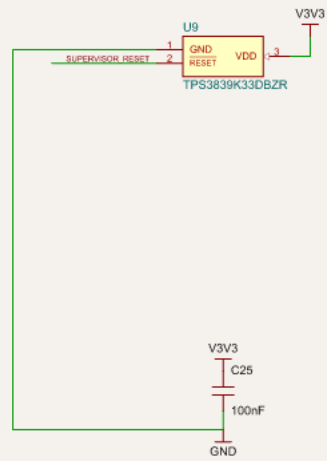
6



U2 / QSPI program flash
16 Mbit (2 MiB); powered from 3.3 V
Device supply range: 2.7-3.6 V

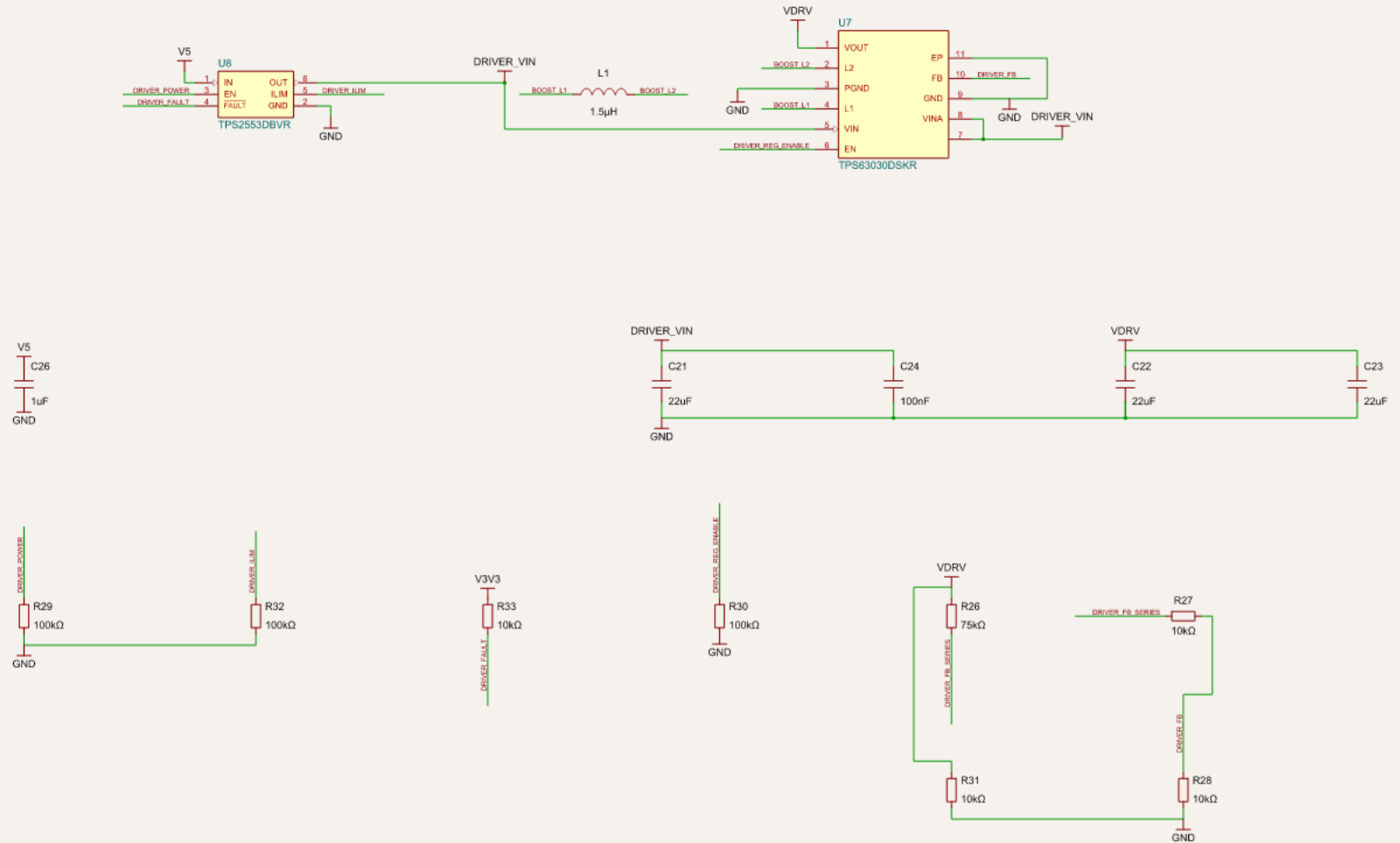


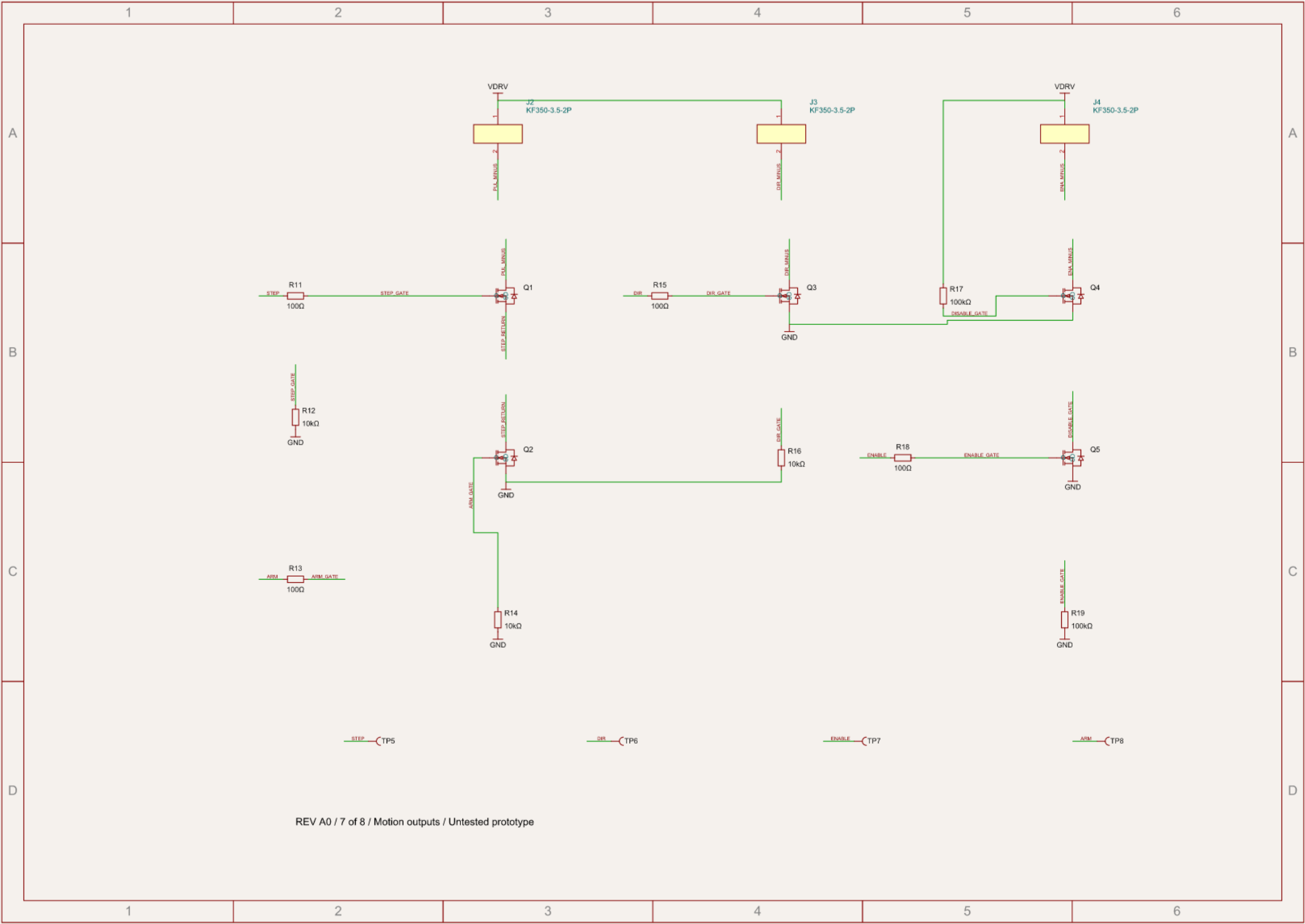
REV A0 / 4 of 8 / Flash and clock / Untested prototype



U8 / switched branch current limiter
 5 V input; ILIM = 100k to ground
 Calculated current limit: 232-306 mA

U7 / buck-boost driver supply
 VDRV = 4.75 V nominal; 48.5 mA budget
 IC input range 1.8-5.5 V; forced PWM





U6 / dual limit-input ESD clamp
3.3 V logic; NC dry contacts only
No 24 V sensors; open wire asserts limit

